JAMES LAVRENZ

Senior Embedded Software Engineer

C/C++ | Embedded Linux / RTOS

AI Inference & Performance Optimization

EMBEDDED SYSTEMS FOUNDATION

Production C/C++ across embedded Linux, RTOS and bare-metal systems. Real-time software, device drivers, DSP, multicore design and hardware/software integration.

HARDWARE-VALIDATED AI RESULTS

Merged ESP-NN acceleration: up to 4.0× on S3 and 40.7× on P4 for batched INT8 FC. A compact-Transformer prototype combining FC and attention reached up to 3.8× end-to-end.

BUILT END-TO-END SYSTEMS

Architected an FP32/INT8 inference runtime for Cortex-M, ESP32 and embedded Linux; designed an FPGA-based 8×8 INT8 systolic GEMM accelerator on Zynq-7020.

ENGINEERING DEPTH

Arm, Xtensa and RISC-V; FreeRTOS, VxWorks and Zephyr; Linux drivers, vector/SIMD optimization, dual-core dispatch, profiling, Git and CI.

INDUSTRY & EDUCATION

Engineering roles at Honeywell Aerospace, General Dynamics and Emerson. M.S. Software Engineering; B.S. Electrical Engineering.

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SELECTED ENGINEERING CONTRIBUTIONS

Upstream code, measured speedups, and hardware-validated prototypes

ESP-NN / PR #46 — MERGED

Batched INT8 fully connected via optimized 1×1 convolution

4.0× S3 / 40.7× P4

Up to; measured on ESP32-S3 / ESP32-P4

ENGINEERING DETAIL

Maps batched FC (weight zero-point 0) onto optimized 1×1 convolution, with dual-core execution on Xtensa and RISC-V.

SCOPE — Memory-safe scratch and compatible fallback paths.

ESP-DL / PR #347 — OPEN PR

AUTO dual-core scheduling for batched GEMM

Up to 1.9× faster

ESP32-S3/P4 · INT8, INT16 and W8A16

ENGINEERING DETAIL

Benchmark-tuned AUTO dispatch engages a second core when beneficial, with synthetic-shape testing and on-device correctness validation.

SCOPE — Scheduling optimization; not a new GEMM kernel.

ESP-DL / PR #346 — OPEN PR

Faster ESP32-S3 INT8 attention query–key path

1.74× faster

60.4 ms → 34.7 ms in measured test

ENGINEERING DETAIL

Reworks a scalar attention query–key fallback using Xtensa SIMD and assembly for faster symmetric INT8 computations.

SCOPE — Query–key path only; shape and alignment dependent.

ESP-NN / ISSUE #54 — PROPOSAL

INT8 BATCH_MATMUL via 1×1 convolution + correction

Up to 3.8× end-to-end

Compact-Transformer FC + attention stress test

ENGINEERING DETAIL

Per-row zero-point correction enables $Q \times K^T$ and attention×V through optimized 1×1 kernels; sandbox operator gains reached 23.4× S3 / 19.1× P4.

SCOPE — Prototype evidence in TFLM sandbox; no production PR.